

## S1 — gain law

The truth is not a member of this one-curve family. Declared truncation, measured offline on  $\bar{w} - \bar{w}_s \in [0.1, 25]$  from `calc_correction_functions`:

$b_{\text{state}} - 1$  : 13.05 % RMS, 15.34 % sup, range  $[+7.79, +15.34]$  %, one-signed, far field  $\rightarrow +1/8$

The far-field limit is exact:  $c_{\perp} \rightarrow 1 + 9/(8\bar{h})$  gives  $b_{\text{state}} \rightarrow 9/8$ . The supremum sits at  $\bar{w} - \bar{w}_s \approx 1.19$ , i.e.  $\bar{h} \approx 2.19$ , inside the canonical oscillation band — the run's most informative segment coincides with the worst point of the truncation.

The exponent 2 is  $1 + 1/p$  at  $p = 1$ ; both asymptotic anchors give  $p = 1$ , so it is anchored rather than chosen. No 8/9 factor is carried (user decision, 2026-08-12), which is why the declared truncation is 13 % and not 0.7 %.

## S3 — what $\delta a$ is, and what the model claims about it

$\delta a$  is not postulated; it is what makes the law-based propagation exact, so its closed form follows by subtraction. Three properties follow from it directly and are used in S8:

- (i)  $\delta a$  is proportional to the increment, hence exactly zero in a hold.
- (ii)  $1/b_{\text{state}} - 1 \in [-0.133, -0.072]$  is one-signed, but  $\delta a$  is NOT:  $\text{sign}(\delta a) = -\text{sign}(\text{increment})$ , so it reverses with the direction of travel. Measured on the canonical scenario,  $\sum_k \delta a = +0.0680$  over the descent,  $-0.0401$  over the ascent, and EXACTLY ZERO over a closed oscillation cycle — both  $\bar{a}'_{\text{true}} d\bar{w}$  and  $(1 - \bar{a}_{\text{true}})^2 d\bar{w}$  are exact differentials in  $\bar{w}$ . The whole disturbance is acquired on the descent; a closed oscillation offers a constant- $\delta a$  model nothing to lock onto.
- (iii)  $(1 - \bar{a}_w)^2$  spans  $0.83 \rightarrow 0.0019$ , so  $\delta a$  varies over three decades.

$\delta a[k + 1] = \delta a[k]$  is therefore a modelling CLAIM, not a consequence.

DECLARED, NOT DERIVED: the step from the differential law of S1 to the difference equation of S4 is forward Euler and is not stated anywhere in the main file. In the height writing that step is an error on top of a correct model; here  $\bar{a}'$  reads the state, so the discretisation IS the model and the estimator is exactly estimating a slightly different family. The flow is autonomous, so an exact one-step map exists in closed form; adopting it would remove the quadrature error identically. Deferred by user decision.

## S6 — the disturbance column, and a path the main file does not name

The disturbance column of  $F_e$  is the constant 1: unlike a multiplicative parameter it does not vanish at a turning point, in a hold, or in the far field.  $F_e(3, 5) = 0$  because the control law is untouched by  $\delta a$ . Separability: a constant  $\delta a$  drives  $e_{\bar{a}_w}$  as a ramp whereas an error in the gain state is a persistent offset.

$y_1$  ALSO CORRECTS THE GAIN. This is implied by  $K = PH^\top S^{-1}$  with  $H_1 = \mathbf{e}_1^\top$ , giving  $K_1(4) = P(4, 1)/S_1$ , and  $P(4, 1)$  is built by  $F_e(3, 4) = -F_{dw}$  and by  $Q_{34}$ . Nothing in the main file is wrong, but the naming (“position measurement” / “gain readout”) has misled every reader including the author. Measured on the canonical scenario:  $K_1(4)$  and  $|P(4, 1)|$  each rise by a factor  $\approx 160$  at motion onset (medians  $1.93 \times 10^{-3} \rightarrow 3.12 \times 10^{-1}$  and  $1.00 \times 10^{-7} \rightarrow 1.57 \times 10^{-5}$ ), while  $K_2(4) \approx 7 \times 10^{-4}$  — three orders of magnitude smaller. A four-arm nested ablation over 8 seeds: removing  $y_2$  does not reduce the across-seed spread; removing  $y_1$ ’s gain correction drops it  $34\times$ ; removing the MA(2) memory changes almost nothing. The spread is motion-triggered, not time-accumulated (a  $4\times$  longer initial hold adds 11 % to the pre-motion spread).

That path is CORRECTLY PRICED:  $K_1(4)$  is within 0.04–16 % of the realised optimum  $E[e \nu]/E[\nu^2]$ . The spread is therefore the information cost of inferring gain from position error, not a defect; removing the path costs 13.6 points of hold level.

## S8 — what the constant claim omits

$Q_{55}^*$  is computable from the S3 closed form, so what the claim omits is priced rather than assumed. Because  $\delta a \propto$  increment,  $Q_{55}^*$  is zero in a hold and grows with commanded speed: a time-invariant container would be the wrong category even at the right magnitude.

FIRST ARM:  $Q_{55} = 0$ . That is the claim “no stochastic driving of  $\delta a$ ”, not the claim that  $\delta a$  is constant, which S3(ii) already refutes. Pre-registered consequence:  $P_{55}$  decreases monotonically and  $\delta \hat{a}$  freezes late in the run; and since the true  $\delta a$  vanishes in a hold while the model keeps injecting  $\delta \hat{a}$ , the final hold should show a spurious ramp of order the descent accumulation.

OPEN ITEM:  $\delta a$  inherits the frequency content of the increment, so on a sinusoidal trajectory it is dominated by the commanded frequency. The internal model principle then says a constant model cannot reject it asymptotically; the derivable alternative is a two-state oscillator at the known commanded frequency. Not implemented.

## Seed and prior

$\hat{w}_0$  is the nominal wall and enters at the seed and nowhere else, because setting  $\hat{a}_w[0]$  IS setting the integration constant.  $P_{45}[0] = 0$  because  $\bar{a}(\bar{w})$  carries no  $\delta a$  in this writing, so  $\partial \bar{a}[0]/\partial \delta a = 0$  and one term is complete. (The sibling `formC_state_da.tex` has a two-term  $P_{44}[0]$  because there  $\delta a$  enters the LAW.)  $P_{\delta a}$  is the only undetermined number; its contract is invariance under a  $10\times$  sweep.

$\text{floor}_a$  as implemented is the supremum over the planned envelope (0.0306, attained at  $\bar{w} \approx 2.41$ ), applied at a seed height where the local shape error is 0.0056. The main file does not specify which of the two is meant. Measured  $\rho = \text{RMS}(e_{\bar{a}})/\sqrt{P_{44}}$  over 8 seeds is 0.15–0.18 before motion and 1.48 at the end of the run.

## Results that decided each arm

Canonical scenario, 8 seeds [7 11 23 42 101 777 27 31], paired (identical draws), overall relative gain-error RMS in percent:

|                                                  | desc peak        | osc RMS         | overall RMS     |
|--------------------------------------------------|------------------|-----------------|-----------------|
| production tier-1 ( $\bar{w}_s$ pinned at truth) | $1.83 \pm 0.72$  | $0.96 \pm 0.62$ | $0.99 \pm 0.61$ |
| production tier-2 ( $\bar{w}_s$ free)            | $4.22 \pm 1.80$  | $1.90 \pm 1.00$ | $2.07 \pm 1.38$ |
| (a) no $\delta a$                                | $15.36 \pm 6.24$ | $4.26 \pm 1.76$ | $4.68 \pm 2.56$ |
| (b) with $\delta a$                              | $17.52 \pm 6.41$ | $7.39 \pm 7.32$ | $7.69 \pm 4.89$ |
| (a) with $\bar{a}'$ from the true height         | $3.23 \pm 0.98$  | $1.58 \pm 1.01$ | $1.87 \pm 0.93$ |

The wall-position oracle is worth  $2.1\times$  on its own (tier-1 versus tier-2), so any comparison against tier-1 overstates the gap.

The true-slope arm improves 8/8 and is statistically indistinguishable from production tier-2 ( $t = -0.48$ ), and better on the descent peak. Hence the derivation, the implementation and the idea of integrating the gain are all sound: the entire gap is that the integrand carries the state's own error, amplified by  $\partial\bar{a}'/\partial\bar{a} = -2(1 - \bar{a})$ . Production also integrates — its predict is  $x_4 + \bar{a}'(\cdot)$  exactly as here — so it does not re-anchor; the only structural difference is where  $\bar{a}'$  is evaluated.

The additive constant disturbance is falsified in four configurations: harmful on canonical with either slope source, harmful on the Menq ramp ( $\bar{h} 6.67 \rightarrow 1.111$  over 10 s, near-wall gate opened), and  $\delta\hat{a}$  never leaves noise ( $-5.4 \times 10^{-8} \pm 5.3 \times 10^{-6}$  against a prior of  $1.53 \times 10^{-4}$ ). Under the true slope there is no model error left to absorb, so  $\delta a$  can only fit noise and its harm there is expected rather than evidence about the concept.

Breaking the feedback by lagging the slope does not work: evaluating  $\bar{a}'$  at the pre-update state changes nothing ( $+0.076$ ,  $t = 0.65$ ) and smoothing is monotonically harmful. So the gap is the BIAS in  $\hat{a}_w$ , not the noise on it.

Menq's forgetting factor (thesis eq. 4.15,  $P = \lambda_f^{-1}[I - KH]P^f$ ) is falsified on 12/12 paired comparisons across two trajectories, two arms and  $\lambda_f \in \{0.9999, 0.999, 0.99\}$ . It does repair the late over-confidence ( $\rho 1.29 \rightarrow 0.79$ ) but pushes the early over-pessimism further down:  $\rho(t)$  here is not uniformly biased, and a uniform-in-time inflation cannot help a deficit whose sign changes with time.

## The anchor, and where the law's error actually lives (2026-08-13)

$b_{\text{state}} \rightarrow 9/8$  is exact in the far field, so  $b = 9/8$  is the asymptotic ANCHOR and  $b = 1$  is the unanchored form. Both the slope and the seed must carry it:  $\int (1 - \bar{a})^2/b d\bar{w}$  integrates to  $\bar{a} = 1 - b/(\bar{w} - \bar{w}_0)$ , and at  $b = 9/8$ ,  $\bar{w}_0 = 0$  that IS the far-field truth. The seed carried  $b = 1$  regardless of the slope until this was fixed (the fix moved the numbers by  $< 0.06$  points; the arms are now consistent).

Four arms, 8 paired seeds, overall relative gain-error RMS:

|                                   | desc peak        | osc RMS         | overall         | hold bias $\pm$ across-seed sd |
|-----------------------------------|------------------|-----------------|-----------------|--------------------------------|
| law at TRUE $\bar{a}$ , $b = 1$   | $12.96 \pm 2.22$ | $6.24 \pm 1.59$ | $8.26 \pm 1.69$ | $-12.17 \pm 2.32$              |
| law at TRUE $\bar{a}$ , $b = 9/8$ | $3.83 \pm 1.14$  | $1.88 \pm 1.17$ | $2.48 \pm 1.29$ | $-3.42 \pm 2.39$               |
| law at EST. $\bar{a}$ , $b = 1$   | $15.36 \pm 6.24$ | $4.26 \pm 1.76$ | $4.94 \pm 2.71$ | $+0.73 \pm 6.88$               |
| law at EST. $\bar{a}$ , $b = 9/8$ | $15.09 \pm 5.69$ | $4.17 \pm 2.32$ | $5.12 \pm 3.32$ | $+3.33 \pm 7.12$               |

Paired: exogenous  $-5.785 \pm 0.587$ ,  $t = -27.9$ , better 8/8. State-fed  $+0.180 \pm 1.366$ ,  $t = +0.37$ , better

3/8. The paired sd bounds the closed-loop effect at roughly  $[-0.8, +1.1]$  points, so it is at least  $5\times$  smaller than the open-loop effect — an interval, not an absence of evidence.

THE DISSOCIATION IS THE RESULT. The law's error lives ENTIRELY in the bias: anchoring removes 72% of it ( $-12.17 \rightarrow -3.42$ ) and leaves the across-seed scatter untouched ( $2.32 \rightarrow 2.39$ ). Feeding the law the state instead removes the bias already ( $+0.73$ ) and pays  $3\times$  the scatter ( $6.88$ ). So there is nothing left for the anchor to fix in the closed loop, and 4.94% is scatter rather than bias.

Hence the paradox that opened this: feeding the law a PERFECT argument makes it WORSE (8.26 against 4.94). The state-fed integrand does not track the law; it tracks the measurements, which are unbiased and noisy. The state writing's own feedback is what discards the law.

Read off the figures (`formC_dist_cmp_anchor_exo_*` against `_anchor_state_*`): the exogenous arms' error is a SMOOTH DETERMINISTIC curve with two valleys locked to the oscillation, reaching  $-13\%$  and settling there, and the anchor scales the whole curve down by  $\approx 3\times$ . The state-fed arms' error is ragged noise about zero with no valleys, and the two columns are visually indistinguishable.

By this project's own no-trade-off rule, a bias-versus-scatter trade set by ONE choice — where  $\bar{a}'$  is evaluated — means the writing is wrong rather than that a compromise must be struck. Production evaluates at the COMMANDED height (`gap_d = w_bar_d - law_ws`), which is exact and free, and gets low bias AND low scatter at 2.07%.